

# Sliding Mode Control with Composite Dynamics for an Inverted Pendulum on a Cart

## 1 Introduction

This document describes a sliding mode controller (SMC) designed for the stabilization of an inverted pendulum on a cart. The controller is based on a composite sliding surface formulation that combines the system states, and it uses a smoothing (saturation) function to mitigate chattering. The controller design includes detailed tuning of various parameters to balance performance, robustness, and chattering reduction.

## 2 Plant Model

The inverted pendulum on a cart is modeled by a linearized state-space representation. The physical parameters are:

- $m_w = 0.432$  kg: Mass of each wheel.
- $m_p = 5.0$  kg: Mass of the pendulum.
- $I_w = 0.5 \times 0.432 \times (0.0726)^2$  kg·m<sup>2</sup>: Moment of inertia of each wheel.
- $I_p = 5 \times (0.4)^2$  kg·m<sup>2</sup>: Moment of inertia of the pendulum.
- $l = 0.4$  m: Length of the pendulum.
- $r = 0.0726$  m: Wheel radius.
- $g = 9.81$  m/s<sup>2</sup>: Gravitational acceleration.

From these, the derived parameters are computed as:

$$\begin{aligned}a &= 2m_w + m_p + \frac{2I_w}{r^2}, \\b &= m_p \cdot l, \\c &= m_p \cdot l^2 + I_p, \\d &= m_p \cdot g \cdot l, \\\Delta &= a \cdot c - b^2.\end{aligned}$$

The linearized state-space model is given by:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{bd}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{ad}{\Delta} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{c}{\Delta} \\ 0 \\ -\frac{b}{\Delta} \end{bmatrix}.$$

The output matrix is defined as:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

### 3 Sliding Mode Controller (SMC) Design

The SMC is designed based on a composite sliding surface defined as a linear combination of the system states:

$$s = \lambda_1 x_1 + x_2 + \lambda_2 x_3 + \lambda_3 x_4,$$

where:

- $x_1$  is the cart position,
- $x_2$  is the cart velocity,
- $x_3$  is the pendulum angle,
- $x_4$  is the pendulum angular velocity.

#### 3.1 Tuning Parameters for SMC

The tuning parameters are:

- $\lambda_1 = 5$ : Gain for the cart position. Increasing this value forces the sliding surface to emphasize reducing position error.
- $\lambda_2 = 20$ : Gain for the pendulum angle. A higher value emphasizes minimizing angular deviations.
- $\lambda_3 = 1.001$ : Gain for the pendulum angular velocity. This affects the damping and speed of the angular response.
- $K = 25$ : Reaching gain for the sliding mode controller. A higher  $K$  forces the state to reach the sliding surface faster but can increase control effort and potential chattering.
- $\phi = 0.05$ : Boundary layer thickness used in the saturation function. A smaller  $\phi$  makes the control law closer to a discontinuous sign function (increasing chattering), while a larger  $\phi$  smooths the control effort.

### 3.2 Effective Gain Computation

For linearized analysis, we form a vector  $L$  for the sliding surface:

$$L = [\lambda_1, 1, \lambda_2, \lambda_3].$$

The equivalent component of the derivative of the sliding surface (without the control term) is:

$$L_{eq} = \left[ 0, \lambda_1, \frac{d}{\Delta}(-b + \lambda_3 a), \lambda_2 \right].$$

An effective gain factor is defined as:

$$k_u = \frac{c - \lambda_3 b}{\Delta}.$$

Thus, the effective state-feedback gain for the SMC is computed as:

$$K_{eff} = \frac{1}{k_u} \left( L_{eq} + \frac{K}{\phi} L \right).$$

### 3.3 Closed-Loop Dynamics

The closed-loop system (linearized equivalent) is then represented as:

$$A_{cl} = A - B K_{eff}.$$

This matrix  $A_{cl}$  is used for further analysis (such as frequency-domain studies) if desired.

## 4 Simulation Using ODE45

The SMC performance is tested by simulating the nonlinear closed-loop system using MATLAB's `ode45`. The simulation is set up with:

- Simulation time span:  $[0, 5]$  seconds.
- Initial state:  $x(0) = [0; 0; 0.1; 0]$  (indicating a small pendulum angle deviation).

### 4.1 Composite Dynamics Function

The function `smc_composite_dynamics` computes the time derivative of the state vector. It performs the following steps:

1. **Unpack Parameters:** Extracts the system parameters and SMC tuning parameters from a structure.
2. **Sliding Surface Evaluation:** Computes the composite sliding surface

$$s = \lambda_1 x_1 + x_2 + \lambda_2 x_3 + \lambda_3 x_4.$$

3. **Saturation Function:** Applies a saturation function to  $s/\phi$  to mitigate chattering:

$$\text{sat}(s/\phi) = \min \left( \max \left( \frac{s}{\phi}, -1 \right), 1 \right).$$

4. **Equivalent Dynamics:** Calculates the equivalent part of  $\dot{s}$  without the control:

$$\dot{s}_{eq} = \lambda_1 x_2 + \lambda_2 x_4 + \frac{d}{\Delta}(-b + \lambda_3 a)x_3.$$

5. **Control Law:** Computes the control input:

$$u = \frac{1}{k_u} (-\dot{s}_{eq} - K \cdot \text{sat}(s/\phi)),$$

where  $k_u$  is the effective gain factor.

6. **State Update:** Finally, the state derivative is computed by:

$$\dot{x} = A x + B u.$$

## 5 Tuning Guidelines

The performance of the SMC controller can be adjusted by tuning the following parameters:

- **Sliding Surface Gains** ( $\lambda_1, \lambda_2, \lambda_3$ ): Increase  $\lambda_1$  to enforce stricter cart position tracking; increase  $\lambda_2$  to improve pendulum angle correction; adjust  $\lambda_3$  to better damp the pendulum angular velocity.
- **Reaching Gain** ( $K$ ): A higher  $K$  leads to a faster convergence to the sliding surface but can increase chattering. Tune  $K$  to balance response speed and smoothness.
- **Boundary Layer Thickness** ( $\phi$ ): Smaller values yield a control law closer to a discontinuous switch (risking chattering), whereas larger values smooth the control action.
- **Effective Gain Factor** ( $k_u$ ): This depends on the plant parameters and  $\lambda_3$ . Adjustments in  $\lambda_3$  affect  $k_u$  and, subsequently, the scaling of the control law.

## 6 Conclusion

The SMC controller designed herein utilizes a composite sliding surface that is a linear combination of the system states. Through the use of a saturation function, the controller achieves robust performance while reducing chattering. Detailed tuning of the sliding surface gains, reaching gain, and boundary layer thickness is critical for achieving the desired stabilization of the inverted pendulum on a cart. The closed-loop dynamics, simulated using `ode45`, confirm the effectiveness of the design under a small initial perturbation.